USN | | | | | | | | | | [24MTEVT112]

FIRST SEMESTER M.Tech EXAMINATION, JANUARY - 2025

ENERGY STORAGE AND MANAGEMENT SYSTEM FOR EV

Time: 3 Hours **Maximum Marks: 100**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
ii. Missing data may be suitably assumed.

			CO's	Blooms Level	Marks
1.	a.	Give the Architecture of Battery Electric Vehicles, hybrid electric vehicles and plugged in hybrid electric vehicle. Discuss the suitability of the above vehicles considering the distance to be travelled (short, medium, long range). Do a critical comparison on the above three types.	CO1	L3	10
	b.	Explain the role of flywheels in the electric vehicle using flywheels, substantiate your answers with proper mathematical expressions.	CO1	L2	10
2.	a.	Discuss on the different types of ratings used for a battery.	CO2	L1	10
	b.	Give the chemical reactions happening in a lead acid battery	CO3	L2	10
3.	a.	Justify the choice of battery for an hybrid electric vehicle, what parameters to be considered? Why?	CO4	L2	08
	b.	Derive the equivalent circuit of the battery hence discuss on its significance.	CO4	L3	12
		OR			
4.	a.	Justify the need for battery modelling, hence discuss on the concept of approximate battery sizing.	CO4	L2	10
	b.	Compare choice of battery selection for a battery operated electric vehicle and plugged in Electric vehicle.	CO4	L2	10
5.	a.	Compare ultra-capacitors with super capacitors.	CO5	L2	08
	b.	Give the basic principles of generation and storage of hydrogen.	CO5	L1	12
		OR			

Jain (Deemed-to-be University) Learning Resource Centre Jain Global Campus Ramanagar-112

6.	**a.**	Discuss on any on fuel cell, its construction and working.	**CO5**	**L2**	**12**
	b.	Discuss on the future of energy storage.	**CO5**	**L2**	**08**
7.	**a.**	Give the classification of different energy management strategies.	**CO6**	**L2**	**10**
	b.	Discuss on the thermal protection of battery.	**CO6**	**L2**	**10**
		OR			
8.	**a.**	Discuss on the various types of EV charging infrastructure.	**CO6**	**L2**	**12**
	b.	Compare the different energy management strategies.	**CO6**	**L2**	**08**